May 28, 2026

Dear Vietnam Summer School of Science (VSSS) 2026,

I did not come to science through a straight path. My academic journey has been more like an adaptive control process: learning from errors, adjusting through feedback, and gradually finding a more stable direction. For this reason, the theme **"Rethinking Science"** is not only an academic topic to me. It reflects the stage I am currently going through: looking back at how I understand science, growing through each research experience, and preparing more seriously for a long-term academic path.

My interest in science and technology started early, through science and technology competitions during my school years. At that time, what attracted me most was the feeling of turning an idea in my mind into a working model in real life. I used to think that doing science mainly meant building, improving, or developing a better product. But the further I went, the more I realized that I needed to answer a deeper question: how can I develop a research topic in a serious, well-grounded way, with a real contribution to the community?

When I entered university and joined a research lab early, I had the chance to work on different problems, from electronics and artificial intelligence to robotics and control. These experiences helped me build technical skills and expand my academic view. However, they also made me question myself: am I really building a strong research ability, or am I only moving through many separate experiences? In my final year, this question became clearer. I realized that a person who wants to follow an academic path in the long term cannot only ask, "What can I do?" They also need to ask deeper questions: What is the meaning of the problem I am working on? Where are its current limits? And do I have enough patience to go deeply into it?

At first, this doubt made me feel uncertain. Later, I understood that it was a necessary signal. In a control system, an error does not only show that the system has not reached the desired state. If it is observed properly, it helps the system adjust itself. For me, the process of rethinking science also began from such errors: the gap between my wish to do research and my real ability, the difference between finishing a project and asking a deep research question, and the tension between being interested in many fields and needing to build a stable academic direction.

To test this direction more seriously, I decided to take a gap year and step outside my familiar environment. I wanted to experience new academic spaces and challenge myself in different research settings. To me, it was not a break from study, but a serious test of my research path.

During my research experience in Taiwan on tool wear prediction in CNC machining, I saw the large gap between a model and a real working system. Machining data is not always clean, measurement signals are not always ideal, and a model is only meaningful when it can help improve product quality, reduce operational risks, lower production costs, and increase system reliability. From that experience, research was no longer just a modeling problem to me. It became an effort to answer clearer questions: What problem does this research solve? Who does it serve? And in what context does it matter?

Later, in Singapore, when I worked on wireless power transfer for underwater autonomous vehicles, I received another important feedback signal. I realized that modern robotic systems cannot be understood from only one single field. An AUV is not only a problem of mechanical design, control, or energy. It is a combination of system design, operating environment, reliability, autonomy, and long-term deployment. The problem of underwater wireless power transfer also showed me a specific technical challenge: how to maintain efficiency and stability in an environment with position errors, weak coupling, and many uncertainties. This experience helped me define my current research interest more clearly: underwater robotics, adaptive control, and wireless power transfer for autonomous systems in complex environments.

Looking back, I am grateful for those unstable periods. They did not make me lose my interest in science. Instead, they forced me to think more seriously about how I do science. I gradually understood that research does not begin with trying to create something new at any cost. It begins with understanding a meaningful problem, seeing the limits of current approaches, asking a clear question, and building a well-grounded research path. For me, this is an important change: from the mindset of doing projects to the mindset of building a research direction.

From this journey, I hope to join VSSS 2026 to place myself in a broader academic community, where I can learn from scientists and from young people who share a serious interest in science. I hope the Summer School will give me the chance to think more deeply about fundamental questions in research, academic responsibility, and the social meaning of science. At the same time, I believe that my research experiences in different environments, together with my process of redefining my own research direction, can help me contribute a practical perspective to the discussions at VSSS.

My conversations with VSSS alumni at NTU showed me that the Summer School offers more than knowledge. It carries an academic spirit that can inspire and influence others in a positive way. I hope to become part of that spirit, continue building my research path seriously, and after the Summer School, share the values of VSSS with young people around me.

Sincerely,

Long Nguyen Khac